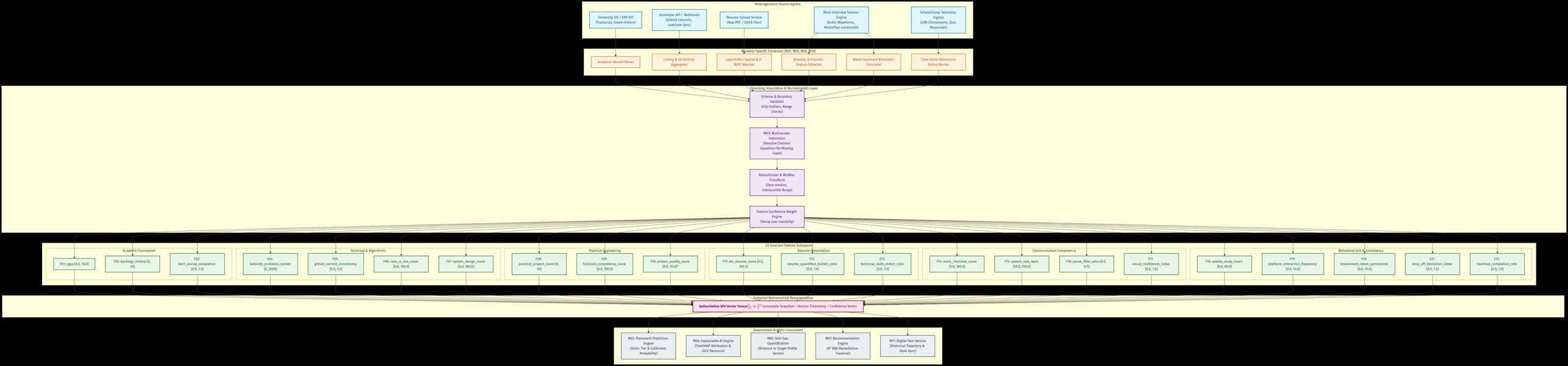

Heterogeneous Source Ingress
University SIS / ERP API (Transcript, Grade History)
Developer API / Webhooks (GitHub Commits, LeetCode Sync)
Resume Upload Service (Raw PDF / DOCX Files)
Mock Interview Session Engine (Audio Waveforms, MediaPipe Landmarks)
ScholarCamp Telemetry Ingress (LMS Clickstreams, Quiz Responses)
Modality-Specific Extractors (M01, M05, M06, M10)
Academic Record Parser
Coding & Git Activity Aggregator
LayoutLMv3 Spatial & S-BERT Matcher
Acoustic & Prosodic Feature Extractor
Wasm Landmark Kinematic Calculator
Time-Series Behavioral Rollup Worker
Cleansing, Imputation & Normalization Layer
Schema & Boundary Validator (Clip Outliers, Range Checks)
MICE Multivariate Imputation (Iterative Chained Equations for Missing Feats)
RobustScaler & MinMax Transform (Zero-median, Interquartile Range)
Feature Confidence Weight Engine (Decay over Inactivity)
22 Invariant Feature Subspaces
Academic Foundation
F01: cgpa [0.0, 10.0]
F02: backlogs_history [0, 20]
F03: tier1_course_completion [0.0, 1.0]
Technical & Algorithmic
F04: leetcode_problems_solved [0, 2000]
F05: github_commit_consistency [0.0, 1.0]
F06: core_cs_dsa_score [0.0, 100.0]
F07: system_design_score [0.0, 100.0]
Practical Engineering
F08: practical_project_count [0, 50]
F09: fullstack_competency_score [0.0, 100.0]
F10: project_quality_score [0.0, 10.0]*
Resume Presentation
F11: ats_resume_score [0.0, 100.0]
F12: resume_quantified_bullets_ratio [0.0, 1.0]
F13: technical_skills_match_ratio [0.0, 1.0]
Communication Competency
F14: mock_interview_score [0.0, 100.0]
F15: speech_rate_wpm [50.0, 250.0]
F16: pause_filler_ratio [0.0, 0.5]
F17: visual_confidence_index [0.0, 1.0]
Behavioral Grit & Consistency
F18: weekly_study_hours [0.0, 80.0]
F19: platform_interaction_frequency [0.0, 14.0]
F20: assessment_retest_persistence [0.0, 10.0]
F21: drop_off_hesitation_index [0.0, 1.0]
F22: roadmap_completion_rate [0.0, 1.0]
Compiled Mathematical Representation
Authoritative SPV Vector Tensor[illegible] ∈ [illegible]^{22} Immutable Snapshot + Version Timestamp + Confidence Vector
Downstream Analytic Consumers
M03: Placement Prediction Engine (Static Tier & Calibrated Probability)
M04: Explainable AI Engine (TreeSHAP Attribution & DiCE Recourse)
M02: Skill Gap Quantification (Distance to Target Profile Vector)
M07: Recommendation Engine (A* DAG Remediation Traversal)
M11: Digital Twin Service (Historical Trajectory & State Sync)